

Machine_Learning_Engineer — Step 5

Learn Data Visualization

Detailed Learning Notes • CareerCompass

1. Learning Objective

Use visualization as an analytical tool to discover patterns, communicate uncertainty and explain model/data behavior without misleading the audience.

2. Core Concepts — Detailed Breakdown

Visual encoding

Understand that position, length and aligned scales are usually easier to compare than area or color intensity. Select charts based on the question, not decoration.

Distribution plots

Use histograms, density plots, box plots and ECDF-style thinking to inspect spread, skew, outliers and multimodality.

Relationships

Use scatter plots, line charts and heatmaps to study relationships, trends and correlation. Always consider whether axes, aggregation or sampling could distort the interpretation.

Categorical comparisons

Use bars for category comparisons and avoid crowded charts. Sort categories when ranking matters and label units clearly.

Analytical storytelling

A good chart has a clear question, useful annotations, appropriate scale and enough context to interpret the result. Avoid unnecessary 3D effects and misleading axes.

Model diagnostics

Visualize residuals, confusion matrices, ROC/precision-recall curves and feature effects where appropriate. Visualization should support model evaluation, not replace it.

3. Recommended Learning Workflow

- Start with a question.
- Choose the simplest chart that answers it.
- Label axes, units and time ranges.
- Check for misleading aggregation or scales.
- Annotate the key insight and state uncertainty/limitations.

5. Hands-on Project / Practice

Create a visualization report for a public dataset containing at least six charts: one distribution, one category comparison, one time trend, one relationship, one correlation view and one model-diagnostic chart. For every chart, write the question it answers and one limitation.

6. Common Mistakes & Pitfalls

- Using too many chart types in one dashboard.
- Truncating axes to exaggerate differences.
- Using color as the only way to encode meaning.
- Plotting raw values when aggregation is required—or vice versa.
- Treating a visual pattern as causal evidence.

7. Completion Checklist

- I can choose a chart from the analytical question.
- I can identify misleading axes and aggregations.
- I can explain distributions and relationships visually.
- I can build readable plots with Python.
- I can communicate one clear insight per visualization.

8. Interview / Assessment Questions

- Why are bar charts useful for categorical comparison?
- When is a scatter plot appropriate?
- How can an axis mislead the audience?
- What is a residual plot used for?
- Why should a visualization include units and context?

9. Recommended References

pandas User Guide: https://pandas.pydata.org/docs/user_guide/

Data Analysis with Python — freeCodeCamp: <https://www.youtube.com/watch?v=GPVsHOIRBBI>

Tableau visualization guidance: <https://www.tableau.com/learn/get-started/desktop-viz-design>

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.